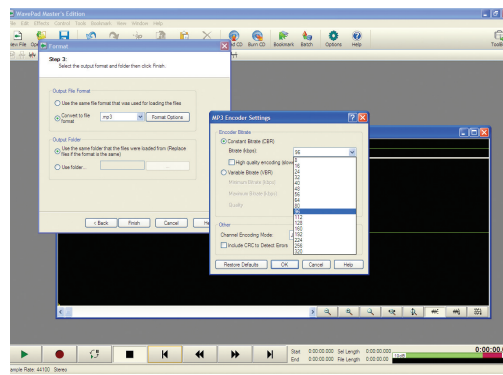


Digital Tools | Tips & Tricks



Set a lower bitrate in the format options

Advanced Noise Reduction

For high quality audio recordings, it is essential to reduce noise for clarity. There are two automatic noise reduction options available, but wavepad also allows a manual approach. Select a section of the audio file that has only noise, the larger this selection, the better the noise reduction will be. Once the selection is made, go to **Effects > Noise Reduction > Grab Noise Sample From Selected Area**. Then go to **Effects > Noise Reduction > Apply Spectral Subtraction Based On Noise Sample**.

Record Audio

Make sure a microphone or an AUX wire is providing input to your computer. Then go to **Control > Record**. Select a sample rate—a sample rate is the number of permanent records made from the signal per second, the default is 44100 hz, which is the same quality of an audio CD. The higher the sample rate, the higher the quality, and the more the disk space used to store the media. Click on the **Stereo** radio button and click on **OK**. A bar showing audio activity should appear. If more than one audio input device is connected to the computer, select the relevant input device in the

device drop down box. If you want wavepad to start recording automatically every time noise is detected, click on **Advanced Recording Options** and check **Auto start recording**. Press the circular red button to start recording.



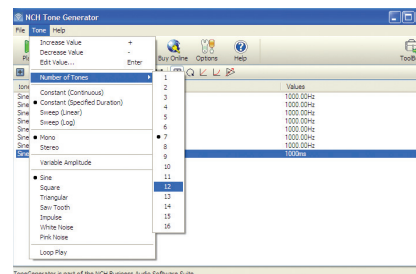
Text To speech

Go to **Tools > Text To Speech**, select a sample rate if prompted, and then type text out in the window and click on **OK**. Wavepad uses Microsoft's text to speech engine, so this feature will not work if you do not have this engine. Every time this is done, a new file with the speech is created. The default is the Microsoft Sam voice, but a few alternatives are available for download from the NCH Web site.

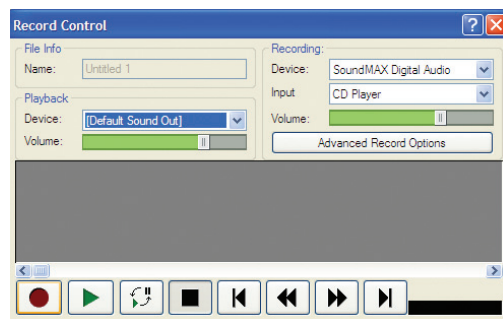


Tone Generator

Go to **Tools > Tone Generator**, which is an install-on-demand module for wavepad. It is a small file, but you will need an Internet connection for this to work. The tone generator has a number of options and can generate sine, triangle, square, sawtooth and impulse train tones. Go to **Tone > Number Of Tones** to change the value from 1 to 16 for the number of parallel tones generated. It is possible to change the frequency of



The tone generator can play upto 16 tones simultaneously



Recording audio using Wavepad

all these tones. Additionally, the tone generator also generates white noise and pink noise. Go to **File > Save As WAV** to save the tone for later use. Clicking on **Edit** will open the tone as a temporary file immediately in wavepad for you to edit.



Remove Clicks And Pops

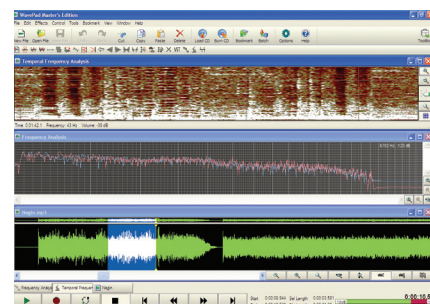
Many recorded tracks have clicks and pops as artefacts of faulty

audio hardware—or when the equipment is moved, handled or toggled. Wavepad can automatically take care of these clicks and pops. Go to **Tools > Auto Click/Pop Restoration**. Maximum click length is the length of the longest defect in the track—the poorer the recording quality, the longer will this be. The default is 450, which should take care of the problem. If the clicks and pops persist, increase it to 600 and run the operation again. If there are still clicks and pops after this, identify the artefact in the waveform, select it and then either press the delete key or go to **Tools > Manual Click/Pop Repair**.



Frequency Analysis

Wavepad offers two kinds of frequency analysis. The basic frequency analysis plots the waveform into its frequency components. For stereo files, there is a blue graph (left channel) and a red graph (right channel). Select a portion of the file, go to **Tools >**



The frequency analysis window in Wavepad

Frequency Analysis. You can zoom in or zoom out in the frequency analysis window to more closely look at the waveform. An alternate mode of frequency analysis offered by wavepad is **Temporal Frequency Analysis**, under **Tools > Temporal Frequency Analysis**. The intensity of any frequency at any point of time is mapped on this graph, giving a hot to cold gradient that can be analysed. A slider adjusts the heat of any particular frequency, which is useful for high intensity music. The brighter areas mean greater intensity (more decibels) of a particular frequency, and the darker areas show a lack of intensity at particular frequencies. This window also has a zoom function, and a grid toggle for reference. ■

aditya.madanapalle@thinkdigit.com